INCH-POUND
MIL-M-38510/360A
AMENDMENT 4
9 January 1998
SUPERSEDING
AMENDMENT 3
31 January 1995

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
ARITHMETIC LOGIC UNITS/FUNCTION GENERATORS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER 8 July 1997

This amendment forms a part of MIL-M-38510/360A, dated 26 August 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute title as printed above.

1.1 Scope: Delete "for each type".

1.2.3 Outline letter: Delete “X” and substitute “2”.

1.3: Delete "Case E - - - - - - - - - - - - - - - 50°C/W
Case F - - - - - - - - - - - - - - - 70°C/W",
and substitute "Cases (all cases) - - - - - - - - - - - - - - - See MIL-M-38510, appendix C".
Delete “Case X” and substitute “Case 2”. After 60°C/W, add “3/”. After Junction temperature (T_J) add "2/".

Bottom of page add new footnotes as follows:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions, as per method 5004 of MIL-STD-883.

“3/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the values stated herein.”

PAGE 2

1.4, After Case operating temperature range: Add “T_C”.

3.2.3 delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

PAGE 3

TABLE I, delete and substitute:

“

				Limits	
				Min	Max
Low-level input current at data inputs	I_{IL1}	V_{CC} = 5.5 V,	All		-.76
0 input	I_{IL2}	V_{IN} = 0.4 V			-.40
Low-level input current	I_{IL3}	V_{CC} = 5.5 V,	All		-.76
Enable input	I_{IL4}	V_{IN} = 0.4 V			-.36

“

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:27Z
Check the source to verify that this is the current version before use.

PAGE 3-continued

TABLE I, Symbol I_{IH2}, Limits max: Delete “200” and substitute “20”. I_{IH3} Limits max: Delete “20” and substitute “200”.

PAGES 4 AND 5

TABLE I, Conditions column: Delete “C_L = 50 pF ±10%” and substitute “C_L = (see figure 5)”.

PAGE 5

* TABLE I, for device 02 only, Symbol t_{PZH}, Limits max: Delete “64” and substitute “76”.

3.6, delete and substitute: “Marking. Marking shall be in accordance with MIL-M-38510.”

PAGE 6

TABLE II, Group A test requirements: Add “8” to class B devices column.

4.2 Screening, c. Delete and substitute:

“c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510”.

PAGE 7

FIGURE 1, Case: Delete “X” and substitute “2”.

PAGE 12

FIGURE 5: Delete and substitute new figure 5 printed on page 3 of this amendment.

FIGURE 5: Delete “(AND E0, GS, OUTPUTS FOR TYPE 02 REGULAR OUTPUT)” under LOAD CIRCUIT FOR DEVICE TYPE 01 as it appears in Amendment 2.

FIGURE 5: Delete “(A0, A1, A2 TRI-STATE OUTPUTS)” and substitute “(ALL OUTPUTS)” under LOAD CIRCUIT FOR DEVICE TYPE 02, as it appears in Amendment 2.

PAGE 13

TABLE III, Symbol column test 16: Delete “I_{IL1}” and substitute “I_{IL3}”. Test 19: Delete “I_{IL2}” and substitute “I_{IL4}”.

TABLE III, I_{IL1} , tests, 11, 12, 13, 14 delete “-.26” min column.

TABLE III, I_{IL2} , test 15 delete “-.16” min column.

TABLE III, I_{IL3} , tests, 16, 17, 18 delete “-.26” min column.

TABLE III, I_{IL4} , test 19 delete “-.12” min column.

PAGES 13, 14, 15, 16, 17, 18, 19, 20 AND 21

TABLE III, Case: Delete “X” and substitute “2”. Delete “Sybmol” and substitute “Symbol”.

PAGE 16

TABLE III, subgroup 10: Delete “T_C = +25°C” and substitute “T_C = 125°C”.

Note 1/: Delete “Case X” and substitute “Case 2”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:27Z
Check the source to verify that this is the current version before use.

PAGE 17

TABLE III, I_{IL1} , tests, 11, 12, 13, 14 delete “-.26” min column.

TABLE III, I_{IL2} , test, 15 delete “-.12” min column.

TABLE III, I_{IL3} , tests, 16, 17, 18 delete “-.26” min column.

TABLE III, I_{IL4} , test, 19 delete “-.12” min column.

PAGE 21

TABLE III, Note 1/: Delete “case X” and substitute “Case 2”.

* TABLE I, for device 02 only, Symbol t_{PZH}, Limits max: Delete “64” and substitute “76”.

* TABLE I, for device 02 only, Subgroup 10: Delete “T_C = +25°C”.

PAGE 22

4.4.3, b.(3): Delete “appendix B of MIL-M-38510 and”.

6.1, Notes: Delete in its entirety.

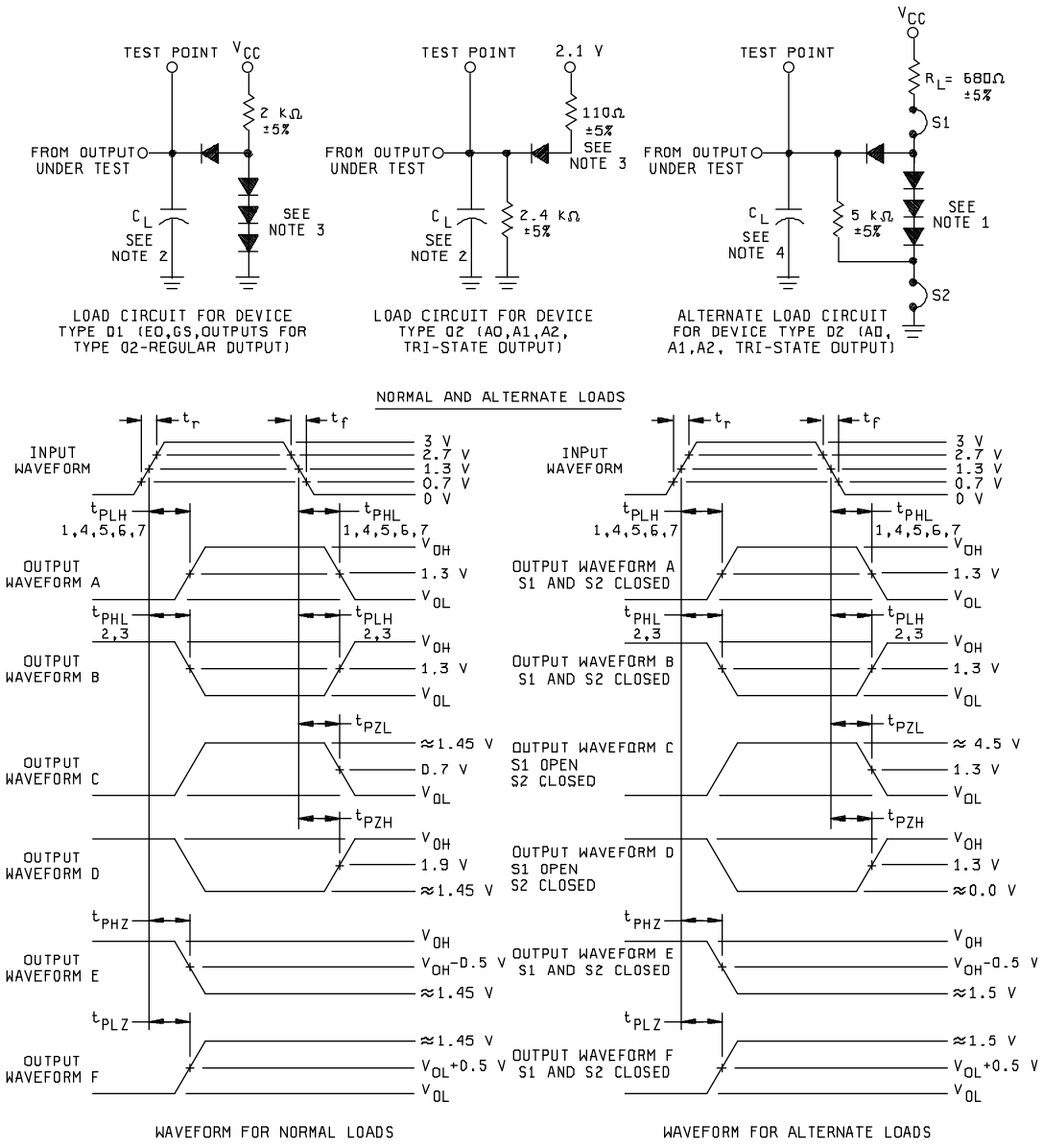
PAGE 23

6.4, GND: Delete “Electrical ground (common terminal)” and substitute “Ground zero voltage potential”.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:27Z
Check the source to verify that this is the current version before use.

*



* NOTES:

1. Input pulse characteristics: PRR ≤1.0 MHz, tr ≤ 15 ns, and tf ≤ 6 ns.
2. For normal load C_L = 50 pF ± 10%.
   C_L includes probe and jig capacitance.
3. All diodes and 1N3064 or 1N916.
4. For alternate load C_L = 50 pF ± 10% for t_{PLH}, t_{PHL}, t_{PZL}, and t_{PZH};
   C_L = 15 pF min. for t_{PHZ} and t_{PLZ}.
   C_L includes probe and jig capacitance.
5. Load circuits on a given output are only required where the specific test given in table III indicates “OUT” on that output. Load circuits may otherwise be omitted.

FIGURE 5. Switching tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:27Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - OS, SH, AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA-CC

(Project 5962-1812)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:27Z
Check the source to verify that this is the current version before use.